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Assignment 2

18. April 2026 — 18. May 2026

Disclaimer: Some questions in this assignment involve reasoning about and implementing code related to language model applications. As with Assignment 1, it is possible that you may find partial or full solutions online or through AI assistants (e.g., ChatGPT). In principle, you could copy solutions directly from such sources. And unless the copying is done sloppily (if we can prove you copied, we will report you to the rectorate) i.e., if you do not take care to cover your tracks, you will likely get away with it. However, we strongly advise against this approach. First, while this may sound cliché, you would only be cheating yourself. ETH Zürich offers you a learning environment guided by experts who are here to support your growth. Take advantage of that opportunity. More practically, the nature of the questions on the assignments is meant to, to a large extent, resemble the possible questions in the final exam and are meant to prepare you for it. Any content covered here is fair game for the exam. Therefore, we strongly recommend that you *think* about the problems *yourself* and try to solve them on your own. You will learn best by doing it.

Collaboration policy: This assignment is *individual* work, and you are required to submit your own solutions. You can *collaborate* with peers in the form of brainstorming, working together, and cross-checking of solutions. However, you must explicitly state who you worked with and, crucially, you must write up your solutions yourself—they cannot be a copy-paste from someone else’s hand-in. The deadline for submitting this assignment is **May 18th 2026, at 23:59**.

Submission instructions: Upload on Moodle a single zip file titled `firstname_lastname.zip`. The zip file should contain one notebook (.ipynb file) for each of the questions.

You can obtain at most 50 points in this assignment The grading scheme is the following:

Assignment 2 grade	Number of points
6.00	[48, 50]
5.75	[46, 48)
5.50	[43, 46)
5.25	[41, 43)
5.00	[36, 41)
4.75	[32, 36)
4.50	[29, 32)
4.25	[25, 29)
4.00	[20, 25)

Question 1: Transformers (20 pts)

In this question, you will implement several core components of the Transformer architecture, the fundamental building block of modern large language models. You will complete the forward passes of key modules, including Layer Normalization, Multi-Head Self-Attention, and the position-wise feed-forward network, and then integrate them into a full Transformer block.

After assembling these components, you will load pre-trained model parameters and use the model to perform autoregressive text generation. This exercise is intended to give you hands-on experience with the internal mechanics of Transformers and to deepen your understanding of how language models process sequences and generate text step by step.

Please see the attached read-only Colab Notebook. Please copy it to your Drive/Colab or download it locally and follow the instructions within. For your solution, submit the filled-in notebook (.ipynb file) named **assignment2_q1.ipynb**. Please **do include** the cell output.

[Colab Notebook Q1](#)

Question 2: Prompting, Calibration and RL (20 pts)

In this assignment, you will investigate how large language models perform on reasoning tasks and how reliable their confidence scores are. You will compare prompting methods, measure accuracy and calibration, and explore self-consistency for improving predictions. Finally, you will apply preference-based fine-tuning and analyze its effect on model performance.

Please see the attached read-only Colab Notebook. Please copy it to your Drive/Colab or download it locally and follow the instructions within. For your solution, submit the filled-in notebook (.ipynb file) named **assignment2_q2.ipynb**. Please **do include** the cell output.

[Colab Notebook Q2](#)

Question 3: Retrieval-Augmented Generation (10 pts)

I do not carry such information in my mind since it is readily available in look up.

-Albert Einstein

And neither should you, that is why our exam allows for cheat sheets. Turns out the principle holds for LLMs too.

Please see the attached read-only Jupyter Notebook. Please download it locally (or reupload to your Drive/Colab) and follow the instructions within. For your submission, submit the filled-in notebook (.ipynb file) named **assignment2_q3.ipynb**. Note: Due to the low total marks, the least count of grading can be 0.1. Please **do include** the cell output.

[Colab Notebook Q3](#)

References. Here's a [simple workflow to get you started with Colab](#):

Colab gives you access to one GPU “for free”, but depending on your usage and on resource availability, the service may decide not to grant you a GPU. We thus recommend that while you're familiarizing yourself with the codebase, or thinking about how to solve the problems, disconnect the GPU runtime. If you have any questions or run into any issues, ask us on Moodle. You can also find good explanations for the things you need to know for the lab (and a lot more) here: [tutorial on numpy](#), [tutorial on pytorch](#), [tutorial on jupyter](#).